CS566 Deep Learning Term Project

Attention-Augmented Scale Hyperprior for Learned Image Compression

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Main takeaway: the SE blocks were easy to integrate and lightweight, but they did not produce a meaningful Kodak rate-distortion gain.

Problem & Motivation

Problem

Images need to be compressed to a low bitrate *without* visibly damaging reconstruction quality.

Why Learned Compression

Neural codecs can learn the encoder, latent prior, and decoder jointly for rate-distortion optimization.

Project Objective

Keep the Ballé scale hyperprior fixed and test whether lightweight SE attention improves compression.

Fair Experimental Question

Compare a fine-tuned vanilla baseline (A_{ft}) against two SE variants: encoder-only (B) and encoder+decoder (C), using **true coded bitrate** on Kodak.

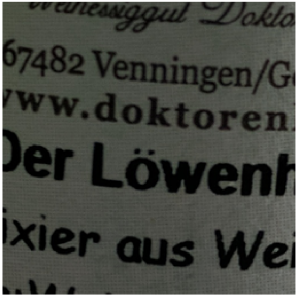
Dataset & Experimental Setup

Final Setup

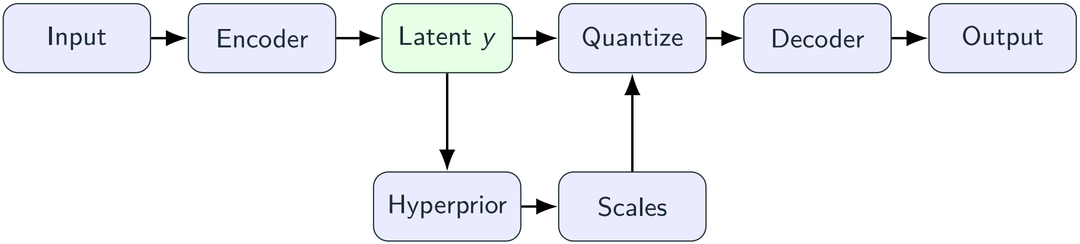
- CLIC train/val: 1,633 / 102
- Kodak test: 24 images
- 256×256 crops; 4 λ values

Example CLIC image

256x256 training patch



Method Overview

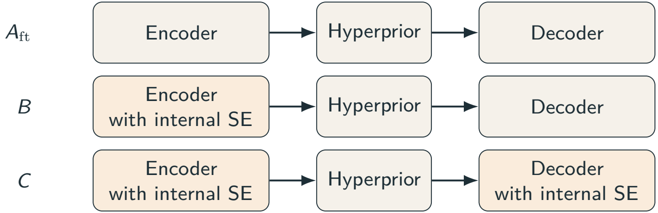


What Changes and What Stays Fixed

Changed: SE blocks are added in the transform network (encoder in B, encoder+decoder in C).

Fixed: the hyperprior, scale prediction, quantization, and entropy model stay the same.

Model Architecture



Implemented Details

- B: SE after encoder conv 2
- C: $+$ decoder SE after deconv 2
- Pretrained transfer; $+2.2\text{k}/+4.4\text{k}$ params

Training Objective

Rate-Distortion Loss

$$\mathcal{L} = \lambda D(x, \hat{x}) + R(\hat{y}) + R(\hat{z})$$

Training Setup

4 operating points

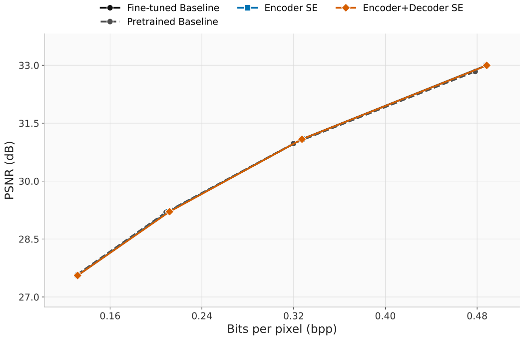
Adam: 10^{-4} backbone, 10^{-3} SE

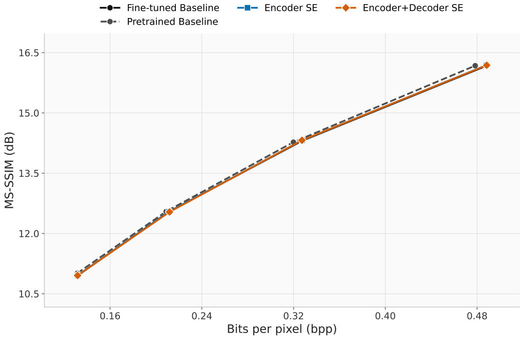
Evaluation

PSNR

true coded bitrate

Results: RD Curves

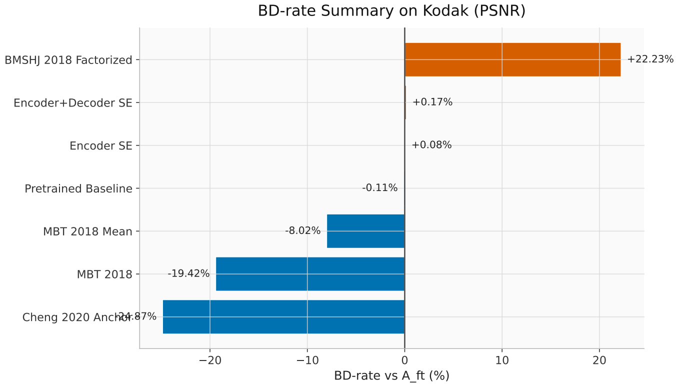




Takeaway

project curves nearly overlap

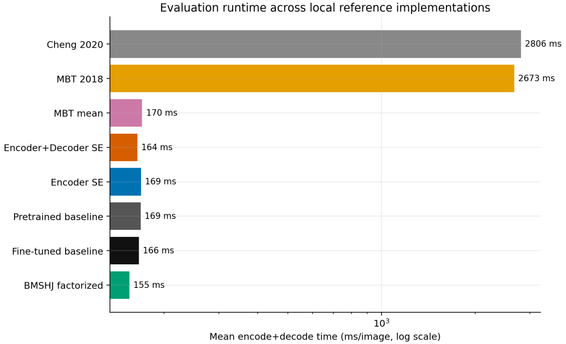
Results: BD-Rate Summary



Takeaway

- B: $+0.08\%$
- C: $+0.17\%$
- Stronger priors: -8% to -25%

Runtime Evaluation



Runtime and Params

Variant	Params (M)	Fwd (ms)	Codec (ms)
A_{ft}	5.1	11.4	166
B	5.1	11.6	169
C	5.1	11.7	164
BMSHJ	3.0	10.4	155
MBT mean	7.0	11.4	170
MBT	14.1	16.0	2673
Cheng	11.833 / 26.599*	26.2	2806

* Cheng 2020 anchor uses 11.833M parameters for qualities 1–3 and 26.599M for quality 4.

Image Comparison

Input | Baseline FT | Encoder SE | Encoder+Decoder SE

no visible difference at quality level 3

Discussion

Why SE Did Not Help Much Here

- **No visual gain:** all variants look the same at quality level 3.
- **Global blanket:** one weight scales an entire channel everywhere.
- **Background bottleneck:** flat regions get boosted along with useful detail, wasting bits.
- **Why Cheng 2020 helps:** spatial attention can suppress flat areas and preserve details.

Conclusion

1. Implemented

Two SE-augmented scale-hyperprior variants, pretrained weight transfer, fair baselines, ablations, and cleaned evaluation.

2. Main Result

On Kodak, the SE variants do **not** show a meaningful rate-distortion gain over the fine-tuned vanilla baseline.

3. Takeaway

Lightweight transform-side attention is easy to add, but stronger priors are the more promising direction.

Thank you

Questions?